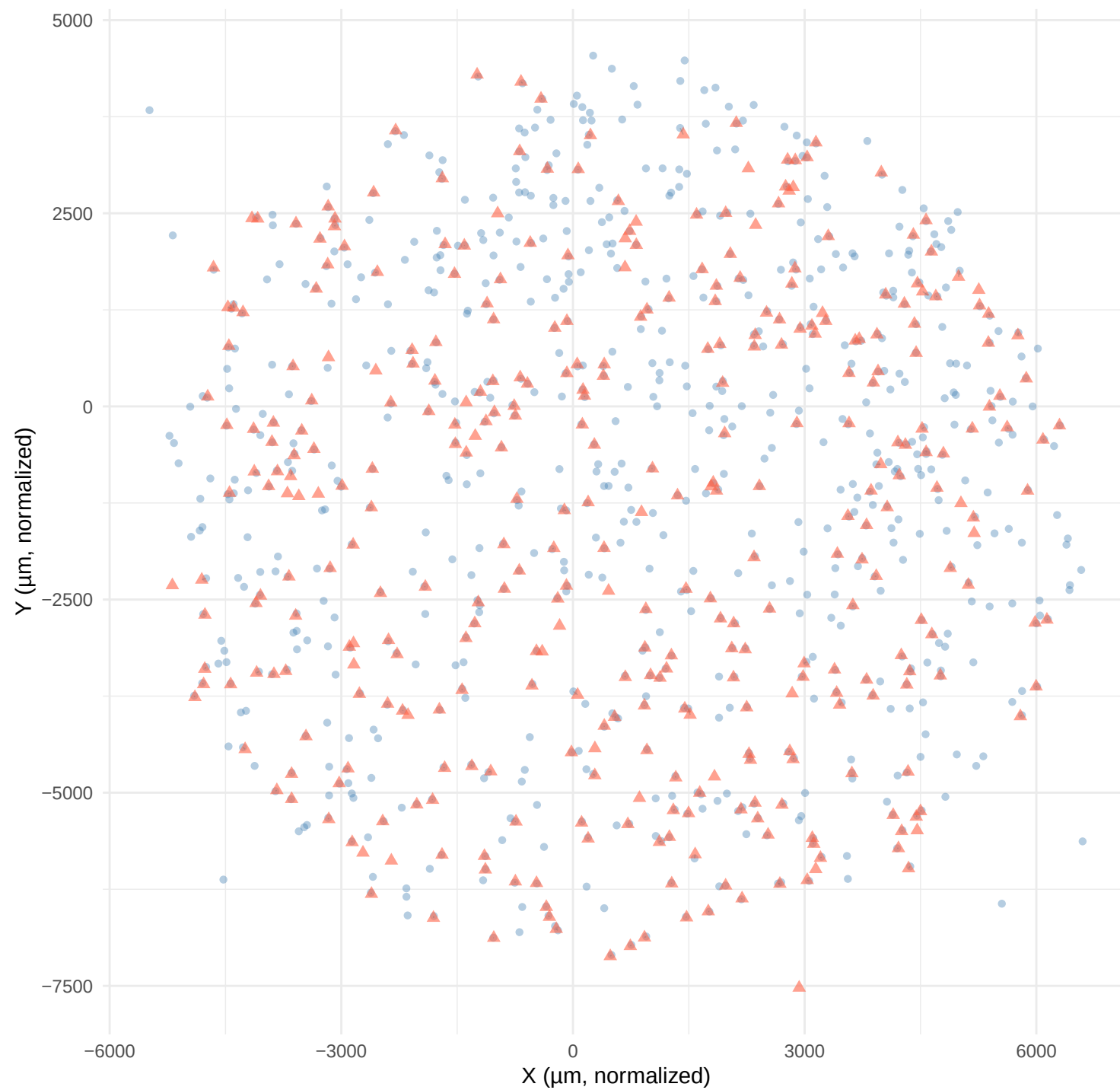


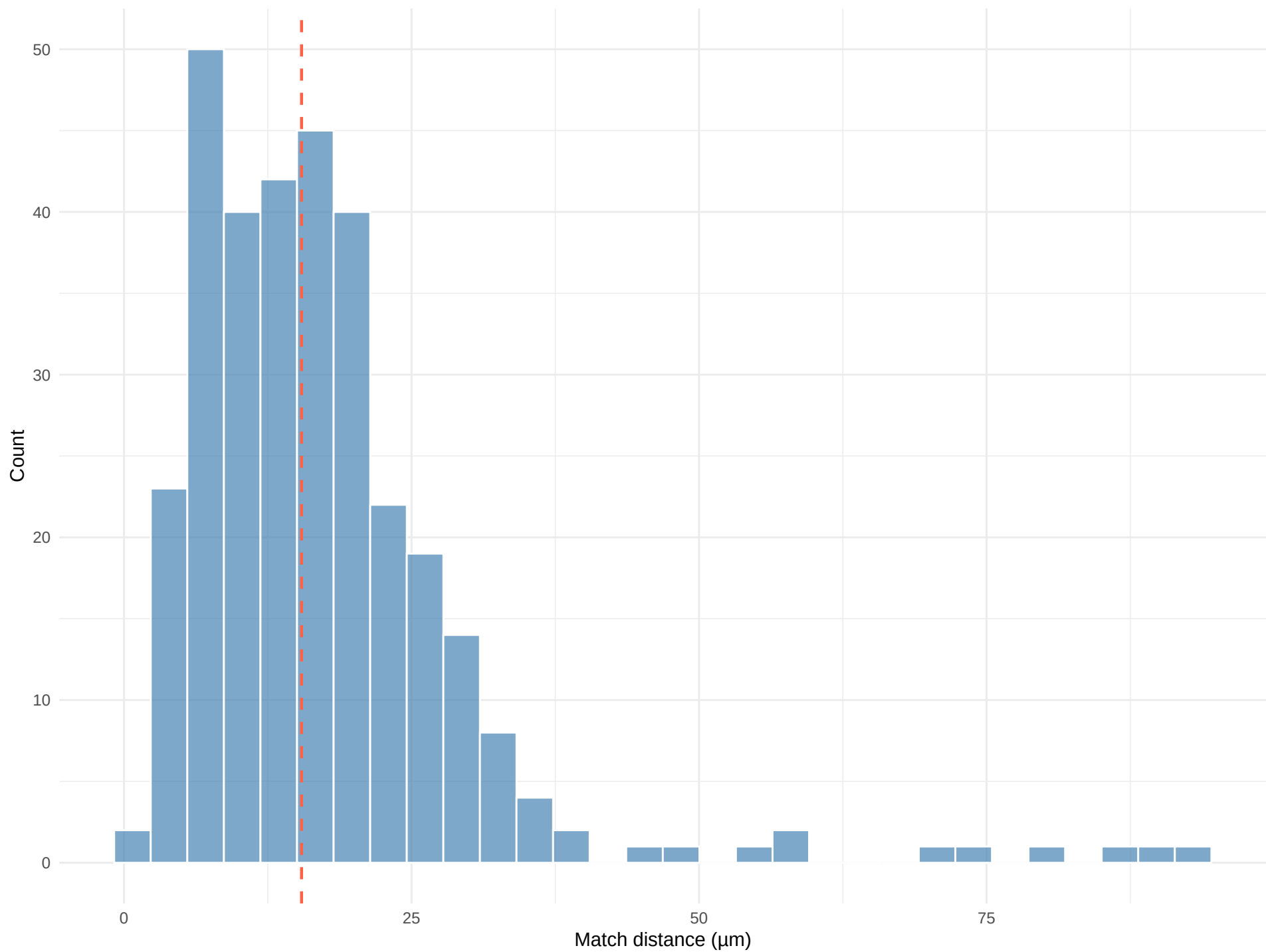
FTIR–Raman Particle Overlay (aligned coordinates)

Red triangles = FTIR, Blue circles = Raman, Lines = matched pairs

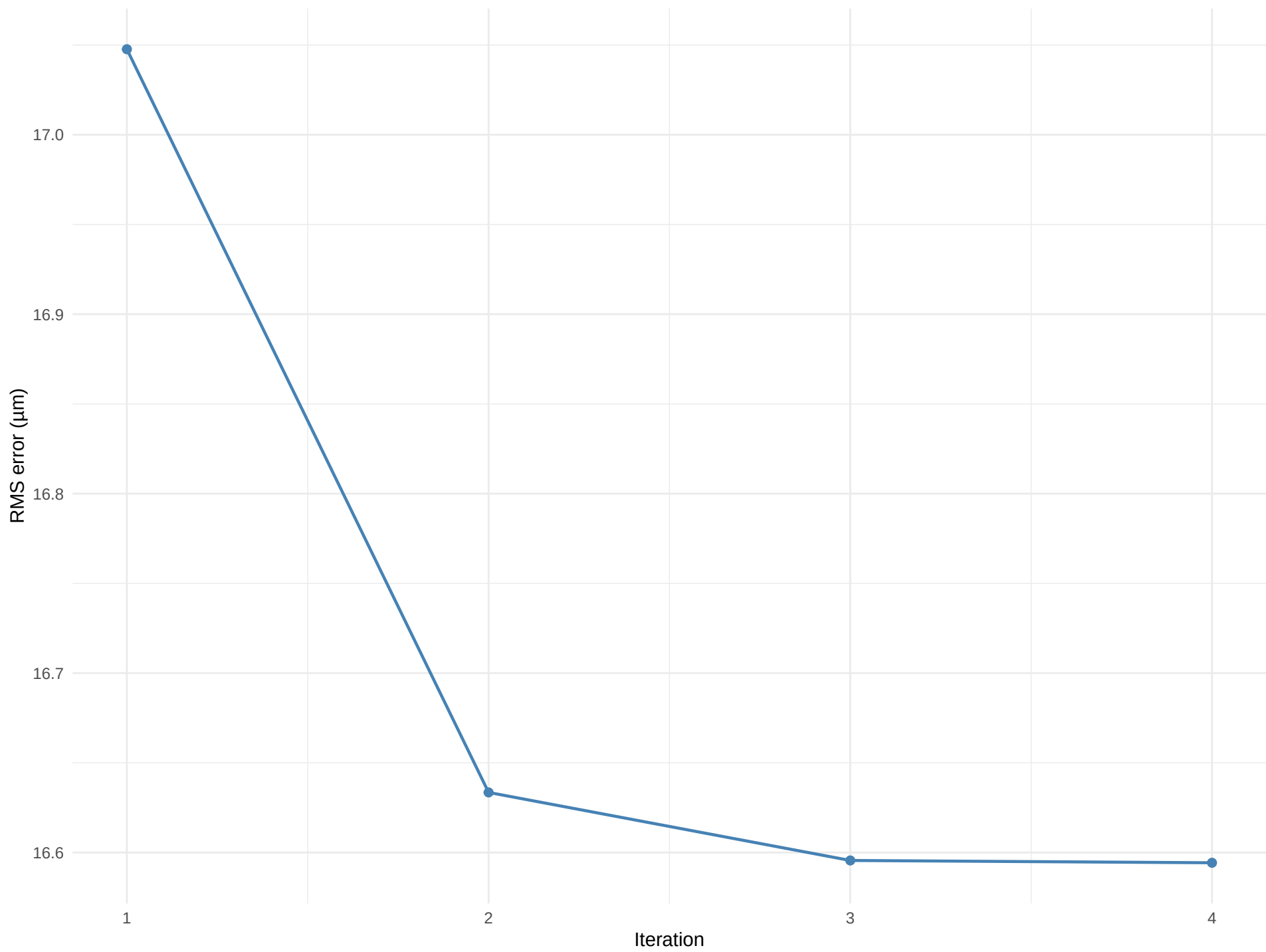


Match Distance Distribution

Median = 15.44 μm , n = 322 pairs

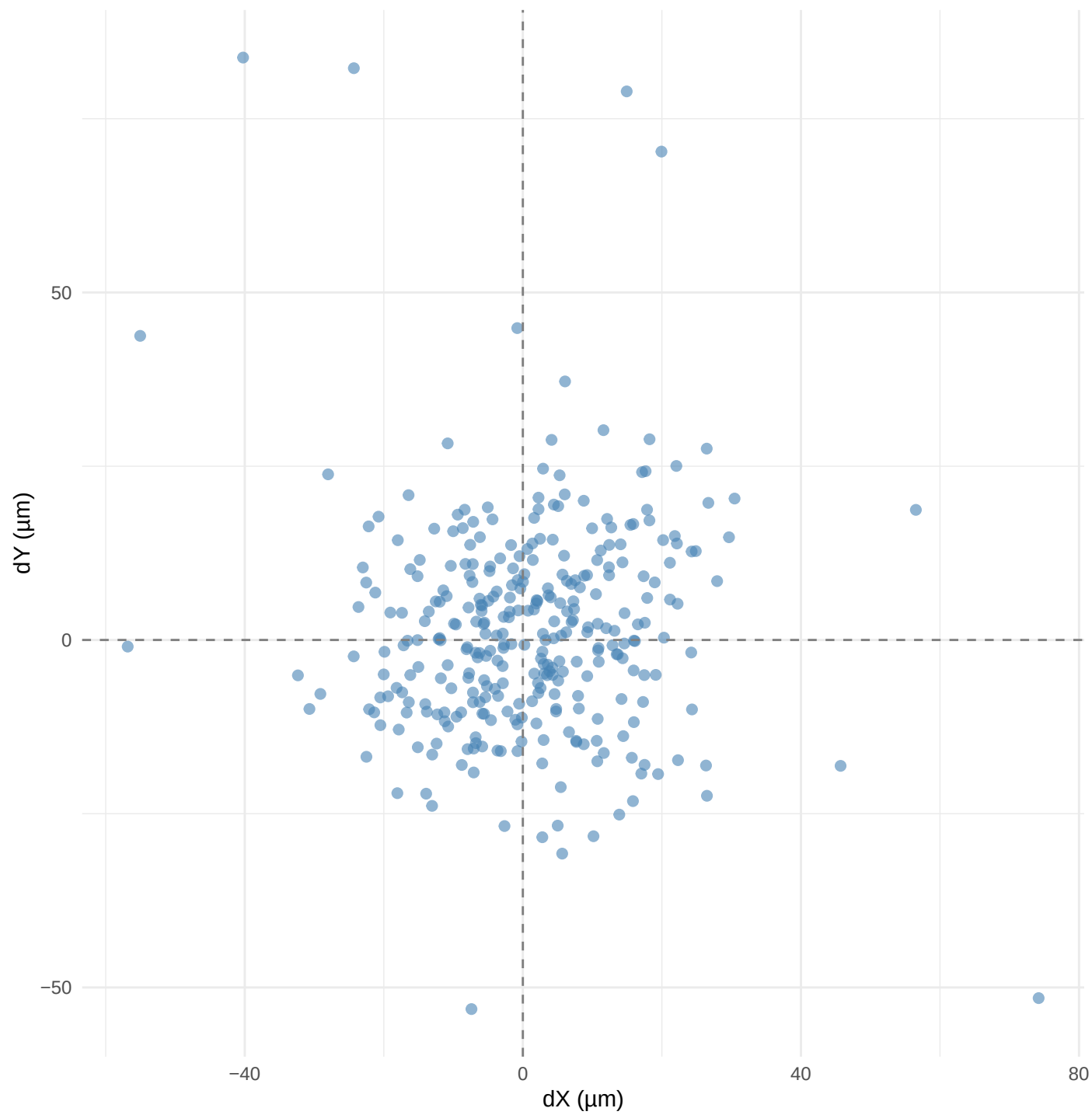


ICP Convergence
Converged: TRUE, Iterations: 4



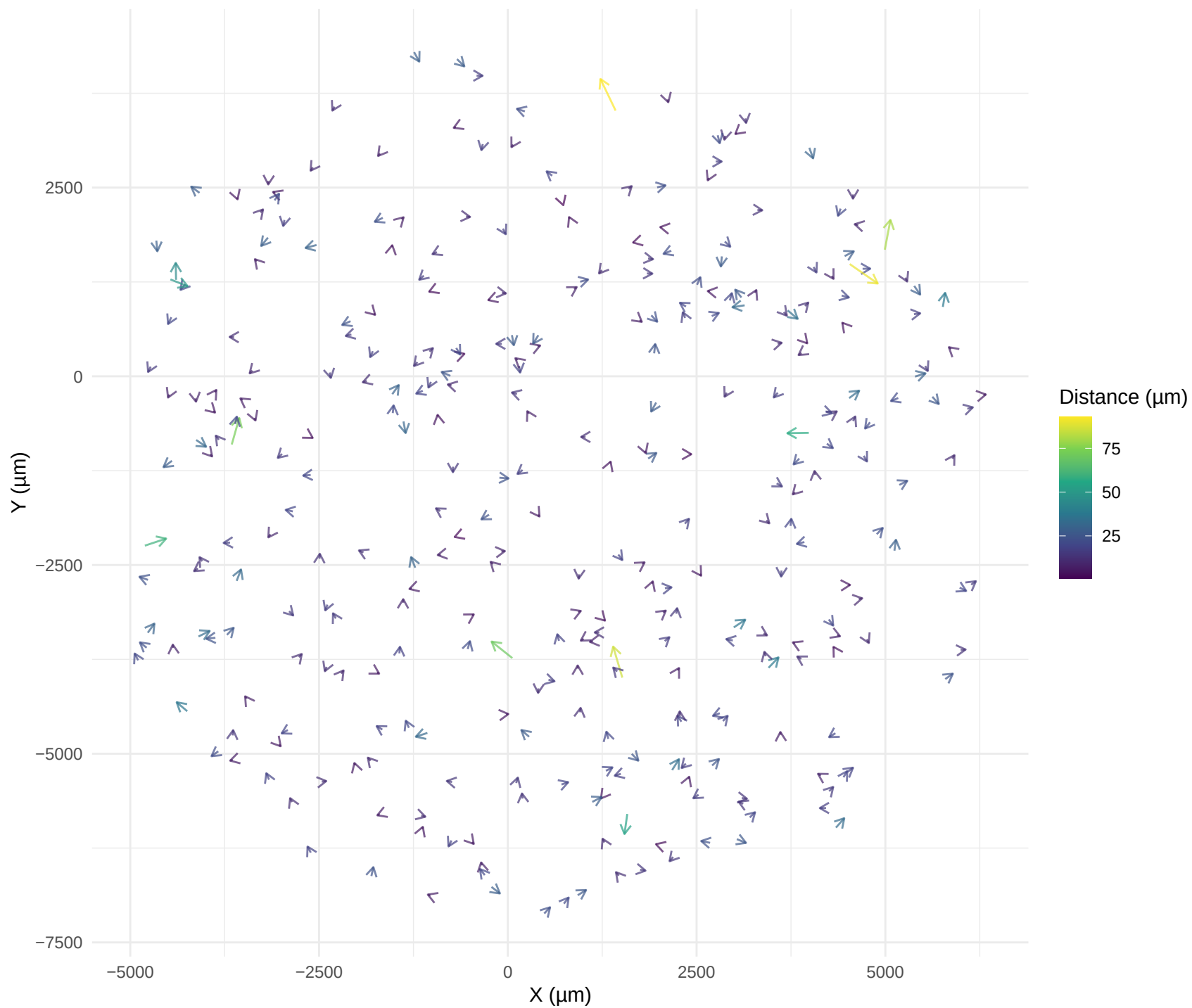
Alignment Residuals

Offset from FTIR (transformed) to Raman (nearest match)



Residual Vector Field

Arrows magnified 5x | Systematic patterns indicate local distortion



Material Confusion Matrix (FTIR vs Raman)

Exact: 0% | Family: 11.5% | Family+: 11.5%

FTIR material

CELLULOSE

PET

PP

PROTEIN

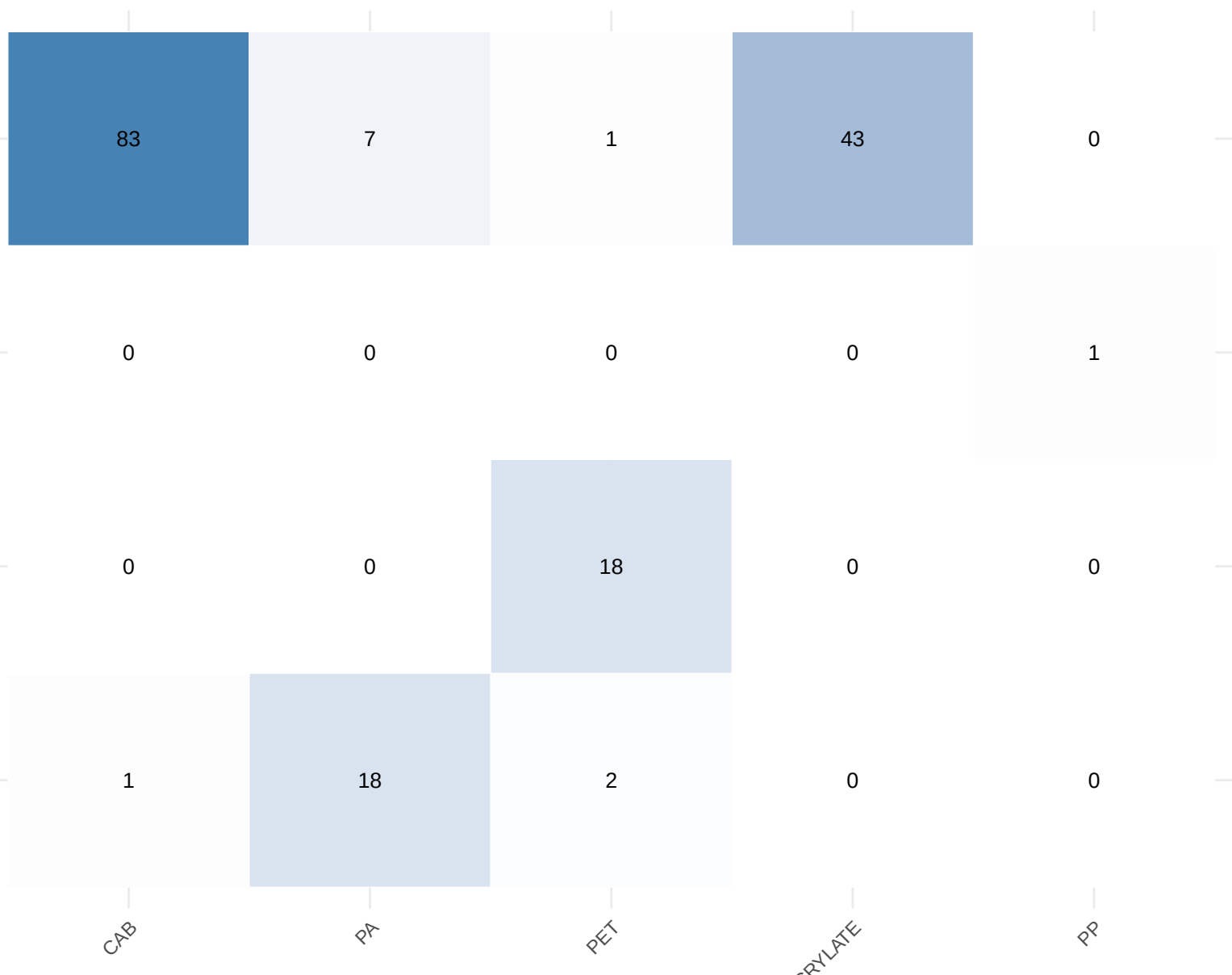
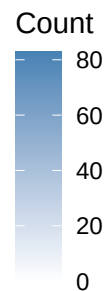
CAB

PA

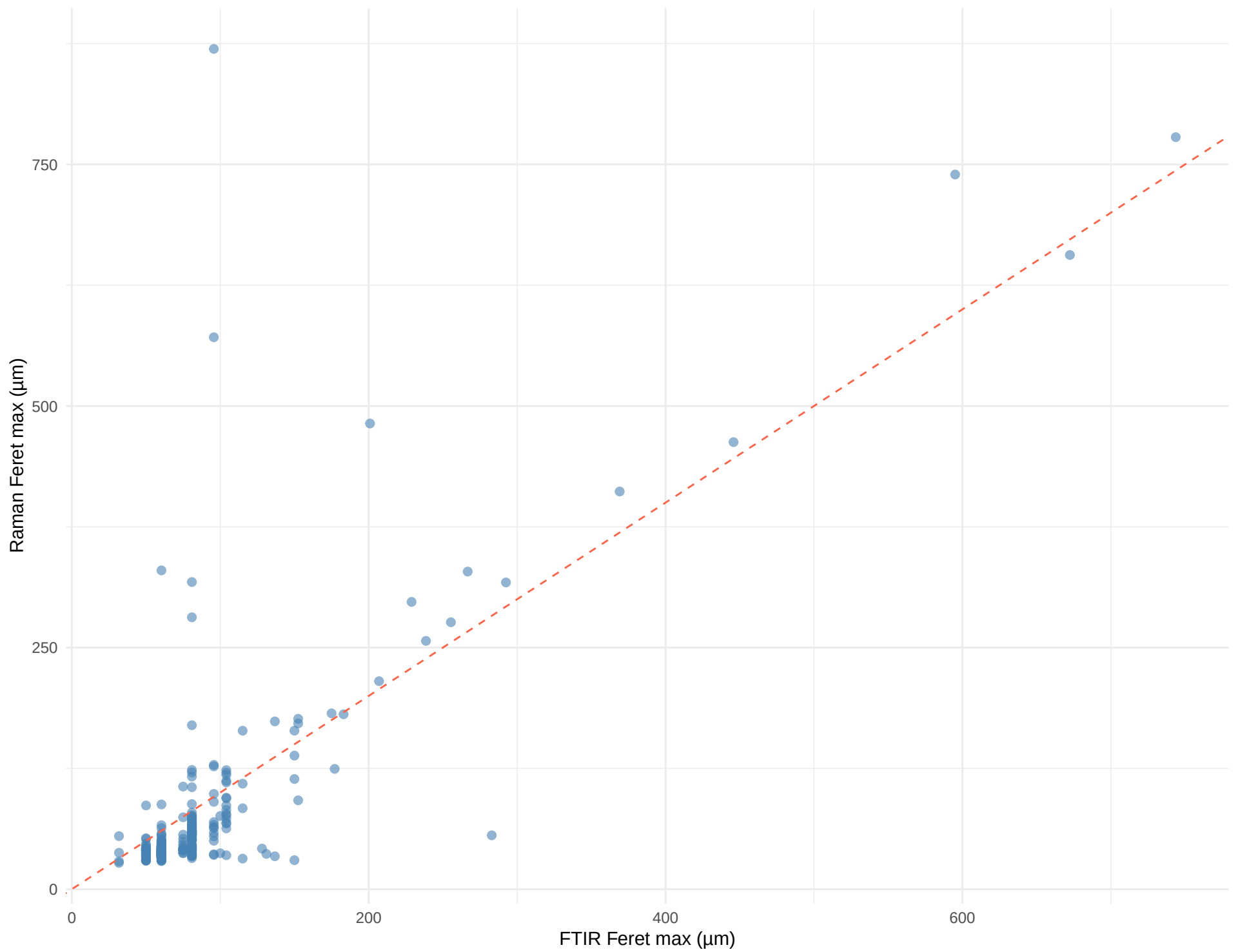
PET

POLYACRYLAMIDE/ACRYLATE

PP

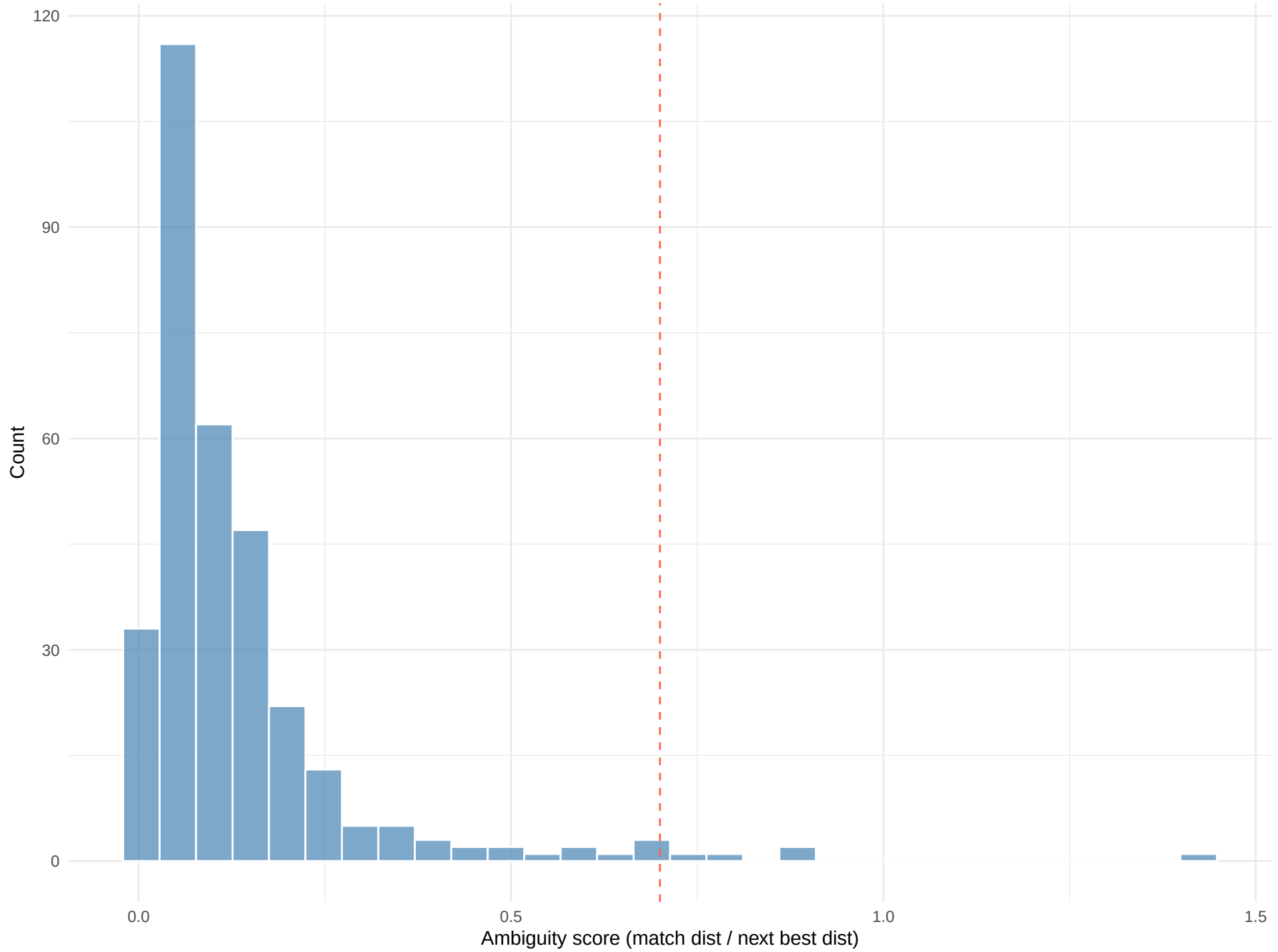


Particle Size Comparison (matched pairs)



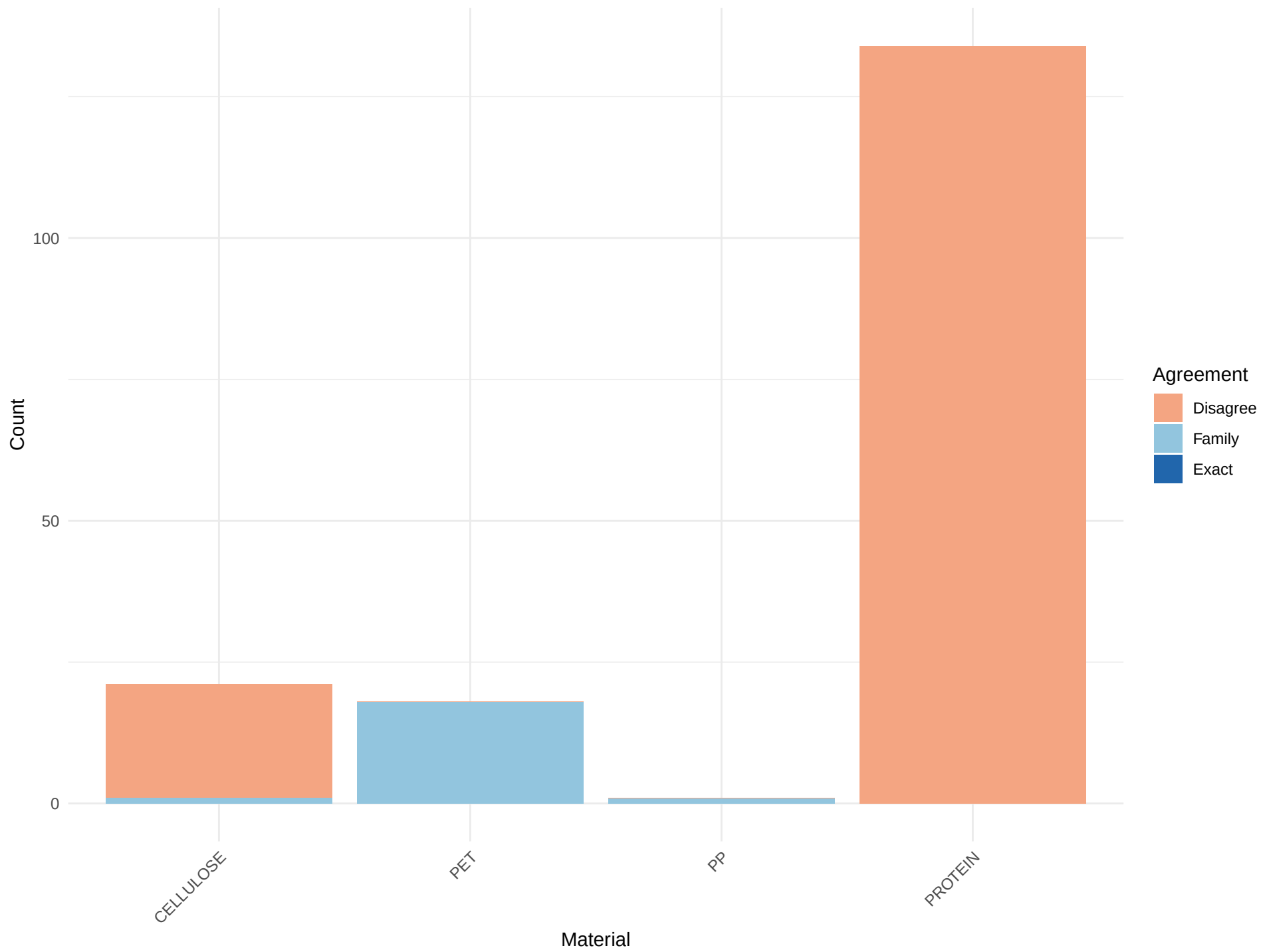
Match Ambiguity Score Distribution

Flagged (>0.7): 7 of 322 pairs



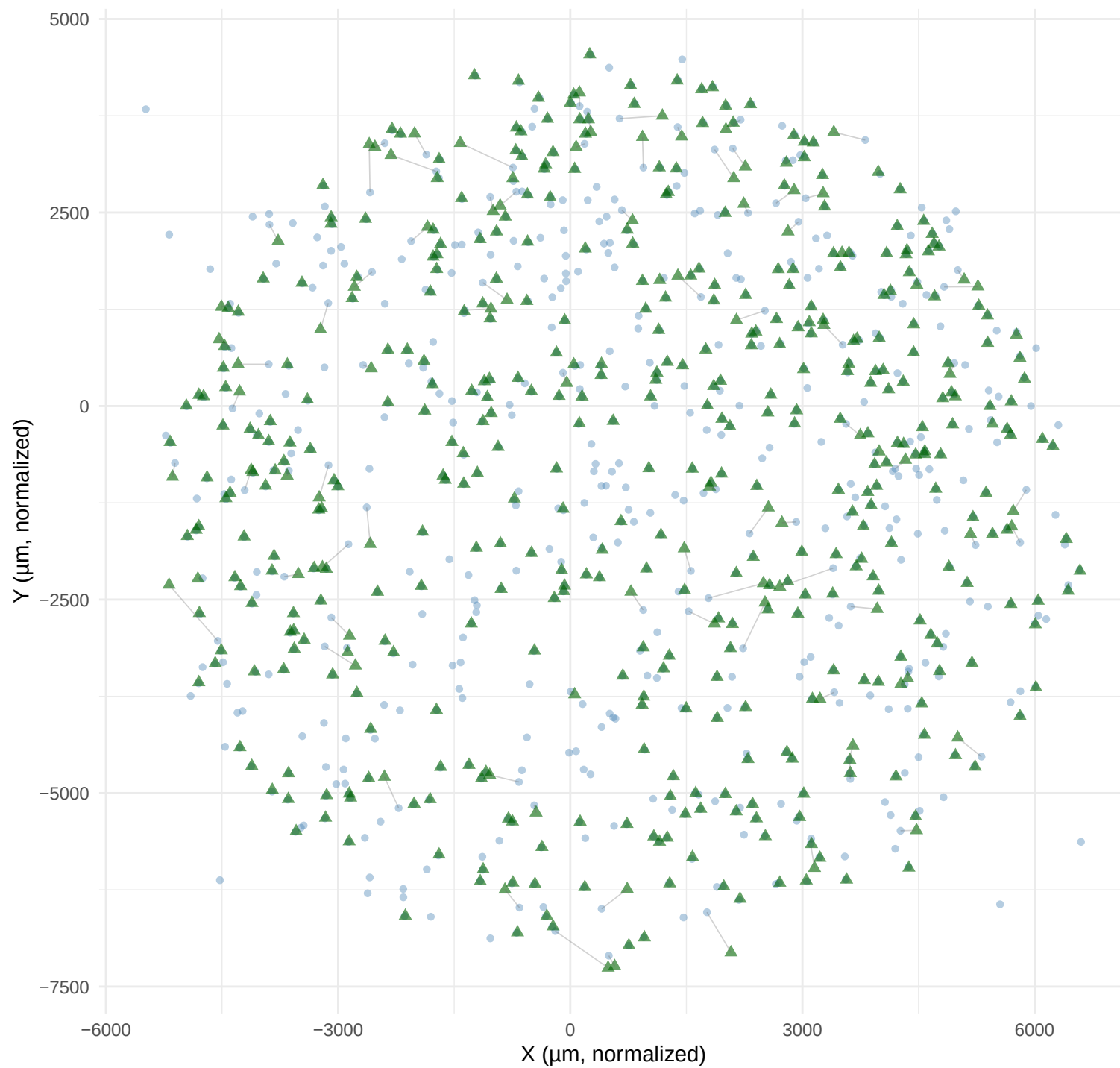
Tiered Agreement by Material

Overall: Exact 0%, Family+ 11.5%



LDIR-Raman Particle Overlay (aligned coordinates)

Green triangles = LDIR (x_aligned/y_aligned), Blue circles = Raman (x_norm/y_norm)



Material Confusion Matrix (LDIR vs Raman)

Exact: 0% | Family: 12.3% | Family+: 12.3%

LDIR material

RUBBER	3	1	0	1	1	0
POLYVINYL CHLORIDE	1	0	0	0	0	0
POLYPROPYLENE	0	1	0	0	0	0
POLYETHYLENE TEREPHTHALATE	9	2	0	20	2	1
POLYCARBONATE	2	2	0	1	0	0
POLYAMIDE	2	1	0	1	1	0
NATURALLY OCCURRING	73	29	1	11	41	1
CHITIN	1	0	0	0	3	0
CELLULOSIC	9	8	0	3	4	0
CARBONATE	1	2	0	0	5	0
	CAB	PA	PE	PET	POLYACRYLAMIDE/ACRYLATE	PP



Raman material